

Jens Laufer

Forward Deployed Engineer — LLM agents running in production, inside the systems a business already has

Python · LLM agents, tool use & MCP · agent harness (loop, memory, approval gates, observability, budget control) · RAG · REST APIs & enterprise integration (SAP, ESB, canonical data models) · Java / Spring Boot · Docker / CI

I build the harness: the loop, the tools, the memory, the gates and the observability that turn a language model into a system somebody can rely on while nobody is watching. Two of mine run in production today — an autonomous AI business partner that plans, writes code, tests and deploys around the clock, and Fabrik, a multi-agent platform where worker agents ship software in parallel. Before that, sixteen years of back-end and integration work sitting with the customer: invoice platforms, handing invoices to SAP over an ESB, master data migration at a major bank, an industry file format in utility billing. That is the forward deployed combination — the AI side, and the side where ERP, CRM and document workflows have been running for decades. Freelance since 2009, so requirements, delivery and production support have always been the same conversation.

AVAILABLE from 15 September 2026	WORKS worldwide	REMOTE SHARE 95 % 5 % on-site
DAY RATE €2,000/day excl. VAT	BASED IN Karlstein am Main Germany	

FOCUS

Harness engineering is the work, in production since 12/2025: agent loop with stop conditions, **tool use over MCP**, layered memory, RAG, approval gates before anything irreversible, quality gates an agent must pass before it may ship, observability over every run, budget and rate-limit control with model routing and fallback, and recovery after a killed run.

Two harnesses running live: an autonomous AI business partner that plans, writes code, tests and deploys around the clock, and **Fabrik**, a multi-agent platform where worker agents pick up tickets and ship in parallel. A local multi-model proxy keeps both independent of any single provider.

Forward deployed by habit: freelance since 2009, always inside the customer's team (Telekom, Commerzbank, Volkswagen, Audi, DVAG, Brunata) — scoping in conversation, delivering on site or remote, staying through production.

Integration into systems that already exist: invoice handover to **SAP** over an ESB (1&1), master data migration on a canonical XML model (Commerzbank), the Haweiko-Arge industry format in heating cost billing (Brunata).

Prototype to production: took over a running prototype at Deutsche Telekom, developed it further and put it live — with the Docker/Kubernetes, tracing and CI work that being live actually requires.

Python and back-end depth: REST back-ends in Python (Eve, Flask) and Java/Spring Boot microservices with OpenAPI, shipped in Docker, test-driven throughout.

ROLES

- Full-stack engineering
- Solution design
- Backend engineering
- DevOps
- Build & configuration management
- Data analysis
- Prototyping
- Machine learning
- Jenkins plugin development
- Maven plugin development

SKILLS

API / INTEGRATION

- REST
- Swagger
- OpenAPI
- SOAP
- Webservices
- CXF
- XFire
- Kafka
- ServiceMix

DEVOPS / CLOUD

- Docker
- Kubernetes
- Rancher
- GitLab CI
- Jenkins
- Maven
- Grunt
- npm
- Artifactory
- AWS (EC2, S3, SageMaker, VPC)
- Microsoft Azure
- DigitalOcean

FRONTEND

- Vue.js
- React
- AngularJS
- GWT
- Quasar
- HTML
- CSS
- Bootstrap

SECURITY / IAM

- Keycloak
- Spring Security
- OAuth2

MACHINE LEARNING

- scikit-learn
- Keras
- DeepLearning4J
- CNN
- Transfer Learning
- Supervised/Unsupervised
- k-means
- PCA
- t-SNE
- Reinforcement Learning
- OpenAI Gym
- Topic Modeling

TOOLING / OTHER

- Git
- GitLab
- Subversion
- Jira
- Confluence
- Kibana
- ANTLR
- JavaCC
- ZXing (QR)
- PayPal API
- RDF/XML/JSON
- SAP (BAPI)
- Phonegap
- JBoss
- Tomcat

LANGUAGES

- Java
- JavaScript
- TypeScript
- Python
- R
- Groovy

MICROSERVICES / BACKEND

- Spring Boot
- Spring
- Spring Data
- Spring Integration
- Spring Security
- OAuth2
- Microservices
- Feign
- Sleuth
- Zipkin
- Python Eve
- Python Flask
- Java EE
- EJB

PERSISTENCE / DATABASES

- MySQL
- MongoDB
- PostgreSQL
- Oracle
- Neo4j
- JPA
- Hibernate

TESTING

- JUnit
- TDD
- Mockito
- Selenium
- DBUnit
- Playwright

DATA SCIENCE

- pandas
- numpy
- dplyr
- Jupyter
- RMarkdown
- ggplot
- seaborn
- Vega-Lite
- A/B Testing
- Webscraping

PROJECT HISTORY

since 12/2025
ongoing

Autonomous AI agent systems — Harness & Agentic Engineering

Solytics GmbH · Remote — AI / SaaS

Harness engineering · Agent engineering · AI system architecture · Full-stack engineering · DevOps

Building and running autonomous LLM agent systems as a **harness engineer** — the engineering around the model, not the model itself. The harness is the product: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **quality gates** that must pass before an agent may ship, **observability** over every run, **budget and rate-limit control** with model routing and fallback, plus recovery when a run is killed mid-flight. A local multi-model proxy keeps the whole thing independent of any single provider. Two systems in production: a self-directing AI business partner that plans, writes code, tests and deploys around the clock, and **Fabrik** (fabrikhq.com), a multi-agent platform for autonomous software delivery with parallel worker agents. Event-driven orchestration over Telegram, GitHub and systemd; several AI-built SaaS products shipped end to end.

Claude / Anthropic APILLM-AgentenAgent HarnessMulti-Agent-OrchestrationTool Use / MCPRAG

Prompt EngineeringPythonFastAPIVue 3GoDockerSystemdGitHub ActionsDigitalOcean

SQLiteTDDTypeScript

07/2024–06/2025
12 months

Digital property profile for owners

verkaufe.immobilien · Remote — Internet

Full-stack engineering · Solution design · DevOps

Full-stack development of a digital property profile that owners create and maintain themselves: Vue.js front-end, Python back-end (Eve) with a REST API and MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development, automated end-to-end tests with Playwright.

PythonPython EveMongoDBTDDVue.jsDockerGitLabGitLab CIRESTPlaywright

07/2022–04/2024
22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry

Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

JavaSpring BootMongoDBJUnitTDDReactDockerGitLabGitLab CISpring Security

KeycloakMockitoOpenAPISwaggerRESTMicroservicesPlaywright

01/2021–03/2021

3 months

Marketing campaign tooling

DigitalMarketer.com · Remote — Marketing

Full-stack engineering · Solution design

Tools that generate marketing campaigns and publish them to Google Ads and Twitter Ads: Java/Spring Boot microservices with REST interfaces (OpenAPI, Feign), Vue.js/Quasar front-end, storage in AWS S3. Designed and built in a microservices architecture. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines. Worked to DevOps principles, owning build and operation of the services in Docker and Kubernetes. Test-driven development (JUnit, Mockito).

Scrum Spring Boot Docker Kubernetes Microservices REST Java OpenAPI Feign JUnit TDD
Mockito Vue.js Quasar AWS S3 GitLab CI GitLab

12/2019–12/2020

13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin. Day-to-day GitLab work (version control, merge-request and review workflow); built and maintained the GitLab CI pipelines for automated build and deployment. Test-driven development (JUnit, Mockito).

Java Spring Boot Docker Kubernetes Microservices REST Spring Data JPA JUnit TDD
Mockito Vue.js MySQL MongoDB HTML CSS GitLab GitLab CI Rancher Kibana Feign
Keycloak Sleuth Zipkin

07/2015–07/2017

25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot Docker Kubernetes Microservices REST Java Maven Spring Security OAuth2
Spring Integration MongoDB Jira Confluence JUnit TDD Mockito Selenium AngularJS Grunt
npm Spring Git

07/2019–12/2019

6 months

Training a stock-trading agent

JiGaoTrading · Remote — Finance

Machine learning · Data analysis

An agent that trades autonomously on Chinese stock exchanges (reinforcement learning).

Reinforcement Learning Python R Docker AWS S3 OpenAI Gym GitLab seaborn AWS SageMaker
MongoDB

04/2010–09/2010
6 months

Web services on an ESB (mediation and rating)

1&1 · Montabaur — Telecommunications
Backend engineering

Software design, web services, maintenance of the rating engine; web services that hand invoices over to SAP.

- Java
- Spring
- Spring Security
- JPA
- Hibernate
- Maven
- JUnit
- TDD
- Mockito
- DBUnit
- ServiceMix
- Webservices
- CXF
- XFire
- SAP-Anbindung
- SAP BAPI
- JBoss
- Oracle
- Subversion (SVN)

05/2009–03/2010
11 months

Master data migration at a major bank

Commerzbank · Frankfurt a. Main — Banking, Finance
Backend engineering

Workflows for persisting master data, a canonical XML model, RDF description for machine processing.

- Java
- Java EE
- IBM MDM (WCC)
- EJB
- JPA
- RDF
- XML
- Oracle
- Subversion (SVN)

CONTACT & CREDENTIALS

CONTACT & PROFILES

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PUBLICATION	Towards Data Science — author

EDUCATION

11/1997	Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent) University of Erlangen-Nuremberg
05/1989	Abitur (university entrance qualification) Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018	Machine Learning Engineer Nanodegree Udacity Verify →
08/2017	Data Analyst Nanodegree Udacity Verify →